Math 118– Lecture 02

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September 3, 2026

1 Norms, Inner Products, Cauchy, etc

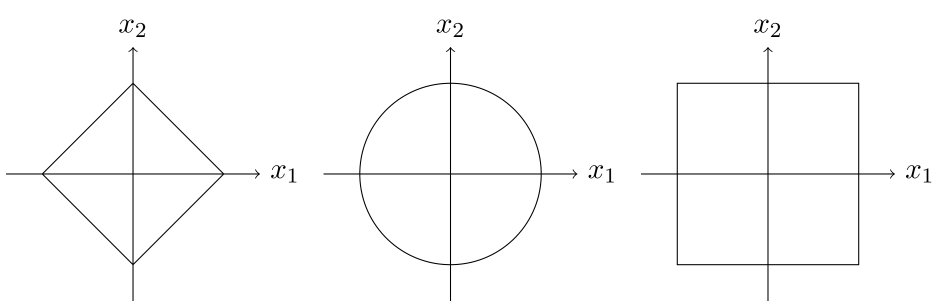
Definition 1.1 (norm)**.** A norm on a vector space V is a mapping $||.|| : V \to [0, \infty)$. with properties

- Triangle Inequality: $||u + v|| \leq ||u|| + ||v||$.
- Homogeneity $||\alpha u|| = |\alpha|||u||$
- Positive Definiteness. $||u|| = 0 \iff u = 0$
- Distance from u and v is $||u - v||$

1.1 Norms on R^n or C^n

- One-norm: $||x||_1 = \sum_{k=1}^{n} |x_k|$
- Two-norm: $||x||_2 = \sqrt{\sum_{k=1}^{n} (x_k)^2}$
- Max-norm: $||x||_\infty = \max_{k \leq n} |x_k|$

Definition 1.2 (Unit Ball)**.** $B = \{x \in V : ||x|| \leq 1\}$



$\|x\|_1 \leq 1$ $\|x\|_2 \leq 1$ $\|x\|_\infty \leq 1$

For any $x \in \mathbb{C}^n$, we have

$$||x||_\infty \leq ||x||_2 \leq ||x||_1$$

because

$$|x||_\infty = \sqrt{\max_k |x_k|^2} \leq \sqrt{\sum_k |x_k|^2}$$

$$x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \qquad e_j = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \quad \text{(1 in the } j\text{th slot)}$$

where $(e_j)_i = S_{ij} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$

So we can rewrite

$$||x||_2 = ||x_1e_1 + x_2e_2 + \cdots + x_ne_n||_2$$

By the triangle inequality we have

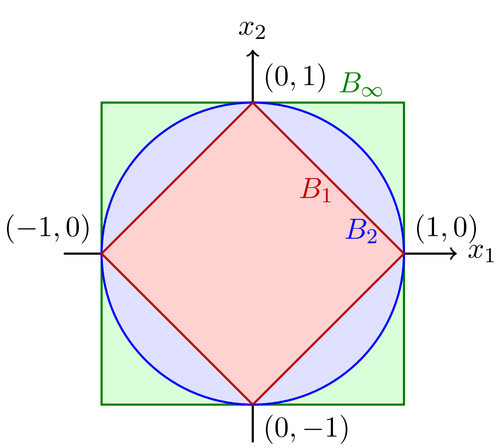
$$||x_1e_1 + x_2e_2 + \cdots + x_ne_n||_2 \leq ||x_1e_1||_2 + ||x_2e_2||_2 + \cdots + ||x_ne_n||_2$$

$$||x_1e_1||_2 + \cdots + ||x_ne_n||_2 = |x_1|||e_1||_2 + |x_2|||e_2||_2 + \cdots + |x_n|||e_n||_2$$

Notice by definition of e_i its two norm would be 1, so this is just $\sum_i |x_k| = ||x||_1$

Remark 1.3 (Unit Balls Nested in the opposite direction)**.** We have

$$B_1 \subseteq B_2 \subseteq B_\infty$$



if $x \in B$, then $||x||_1 \leq 1$ so

$$||x||_2 \leq ||x||_1 \leq 1$$

2 Inner Products

An inner product is a mapping $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{C}$ such that

1. $\langle u, u \rangle > u$ if $x \neq 0$
2. $\langle v, u \rangle = \overline{\langle u, v \rangle}$
3. $\langle \alpha u, v \rangle = \alpha \langle u, v \rangle$.
4. $\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$
5. 3 implies that $\langle 0, v \rangle = 0$
6. 2 and 3 imply that $\langle u, \alpha v \rangle = \overline{\alpha} \langle u, v \rangle$
7. 2 and 4 imply that $\langle w, u + v \rangle = \langle w, u \rangle + \langle w, v \rangle$

Geometrically:

$$\langle u, v \rangle = ||u|| \cdot ||v|| \cdot \cos \theta$$

$$||\alpha u|| = \sqrt{\langle \alpha u, \alpha u \rangle} = \sqrt{\alpha \overline{\alpha} \langle u, u \rangle} = |\alpha| \sqrt{\langle u, u \rangle}$$

In physics,usually the innter product is defined to be by linear in y.

Theorem 2.1 (Cauchy Schwartz) *Prove that*

$$|\langle u, v \rangle| \leq ||u|| \cdot ||v||$$

Proof. □

3 Definitions and notation

Definition 3.1 (First definition)**.** Write the definition and one sentence explaining why it matters.

4 Claims, theorems, and proof sketches

Claim. State the claim in one precise sentence.

Proof sketch. Record the key idea, the first useful equation, and the step that still needs checking.

5 Examples and calculations

Example 5.1 (Lecture example)**.** Write the setup, the calculation, and the conclusion.

6 Questions and follow-up

- What is still unclear?

- Which exercise or reading should I revisit?
- TODO: